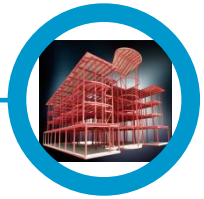
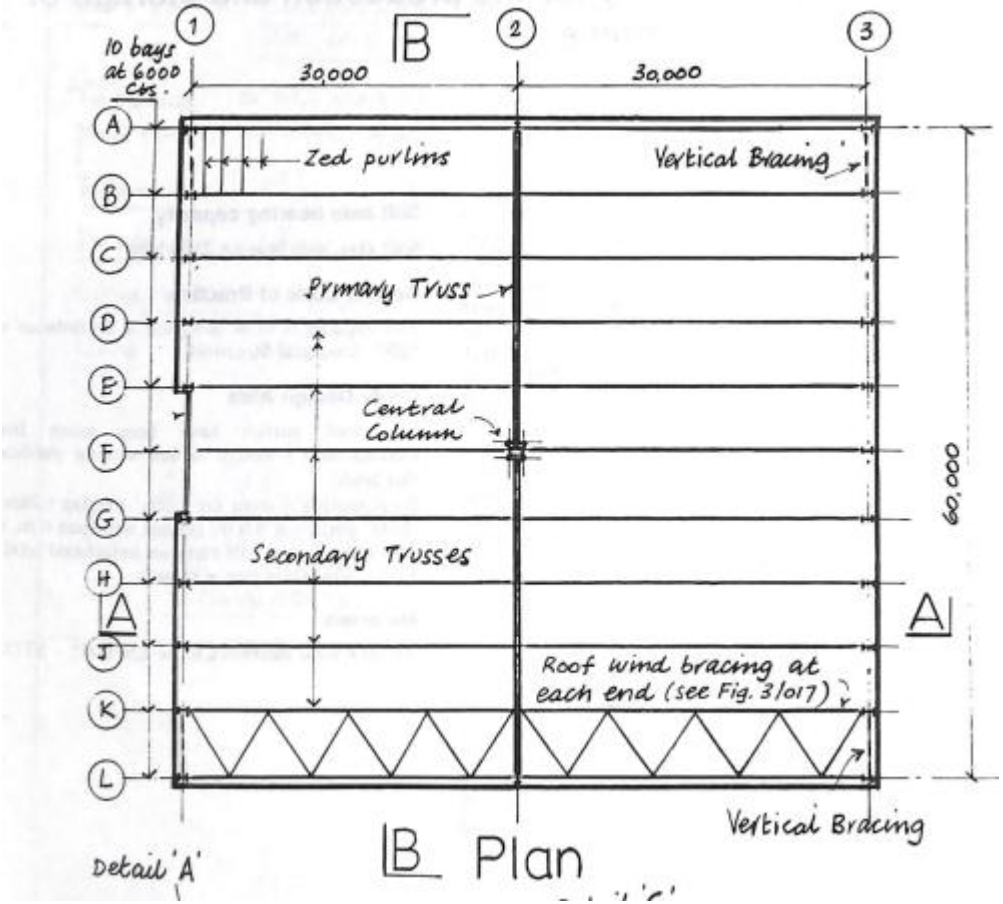


Steel Buildings: Complex Engineering Prob

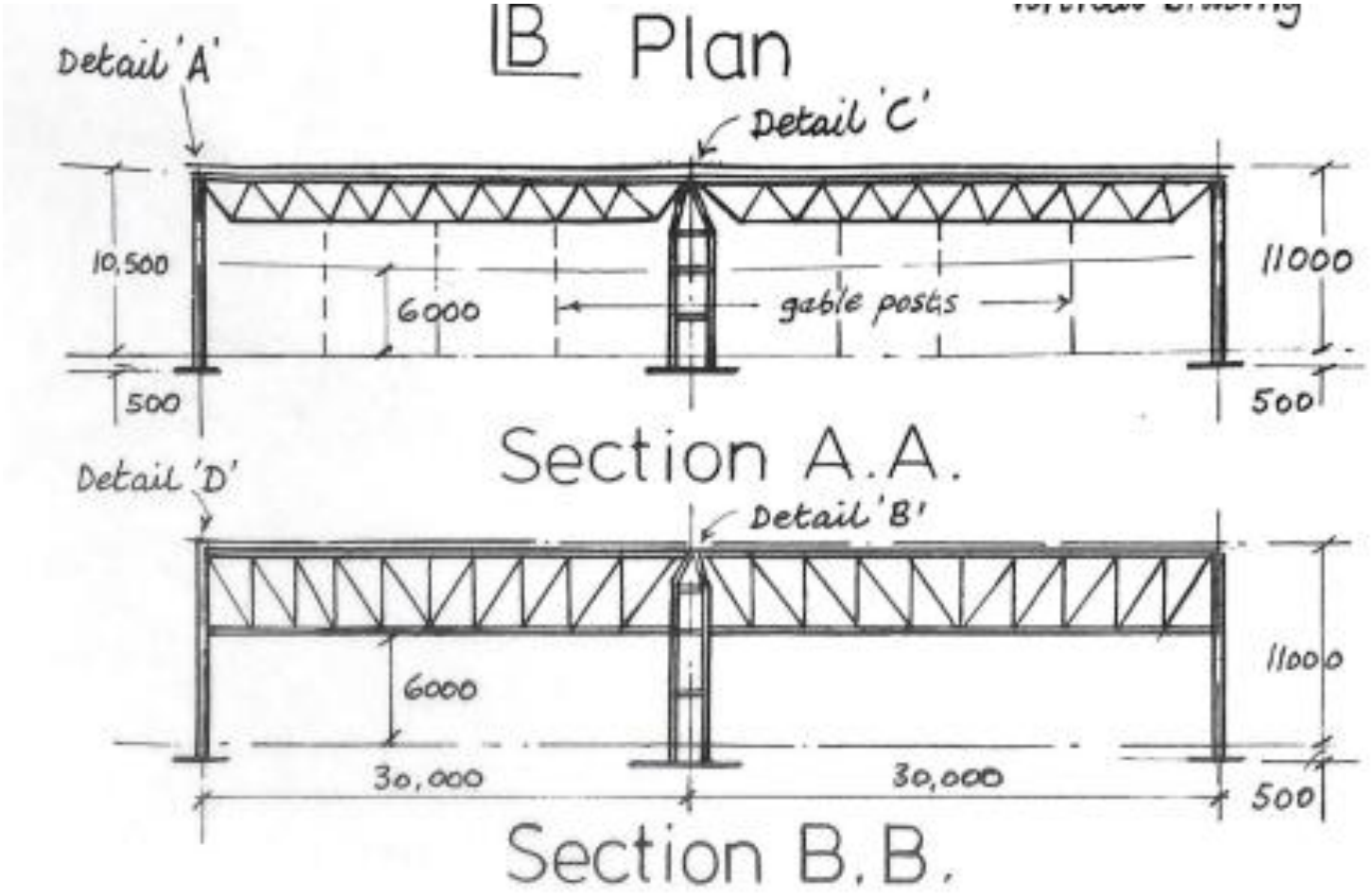


Scheme 1:



Steel Buildings: Complex Engineering Prob

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Steel Buildings: Complex Engineering Prob

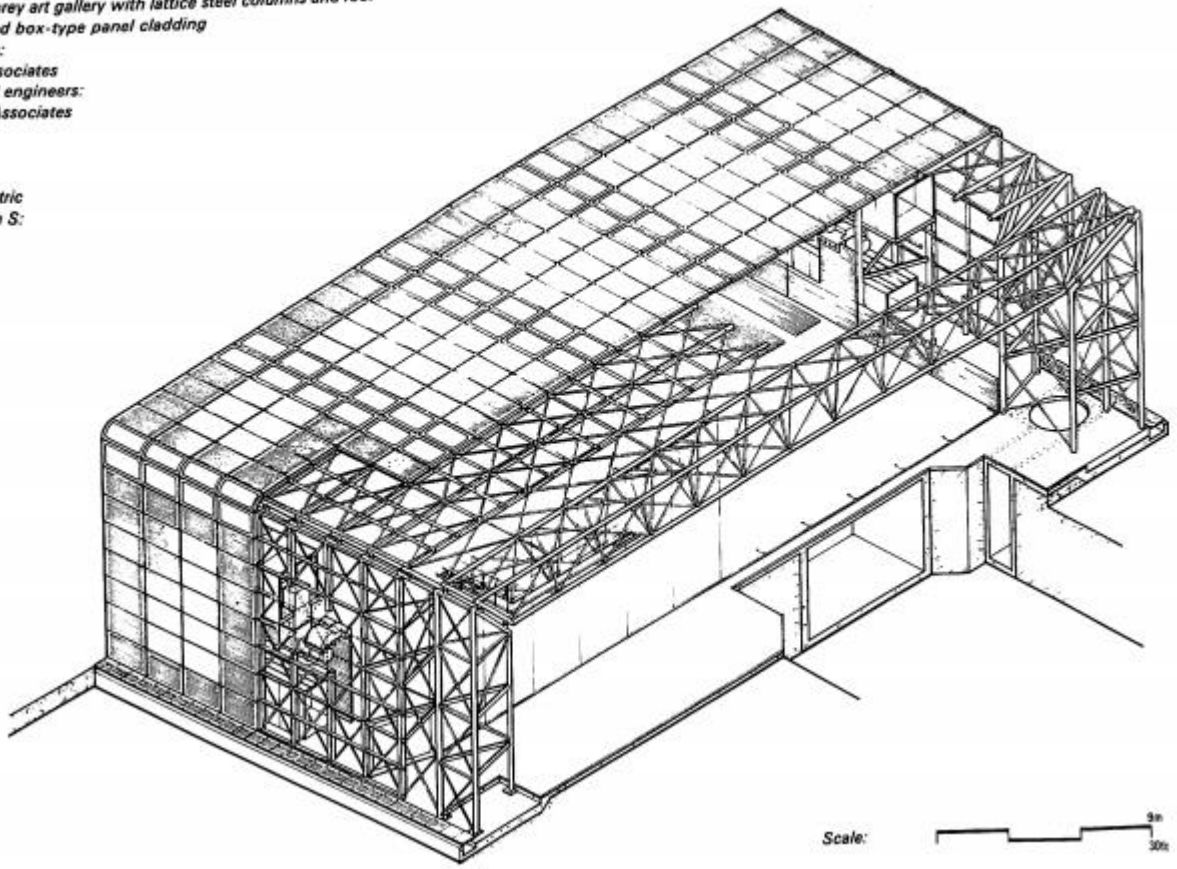
Scheme 2:

Highly Serviced Buildings

SAINSBURY CENTRE (1978)

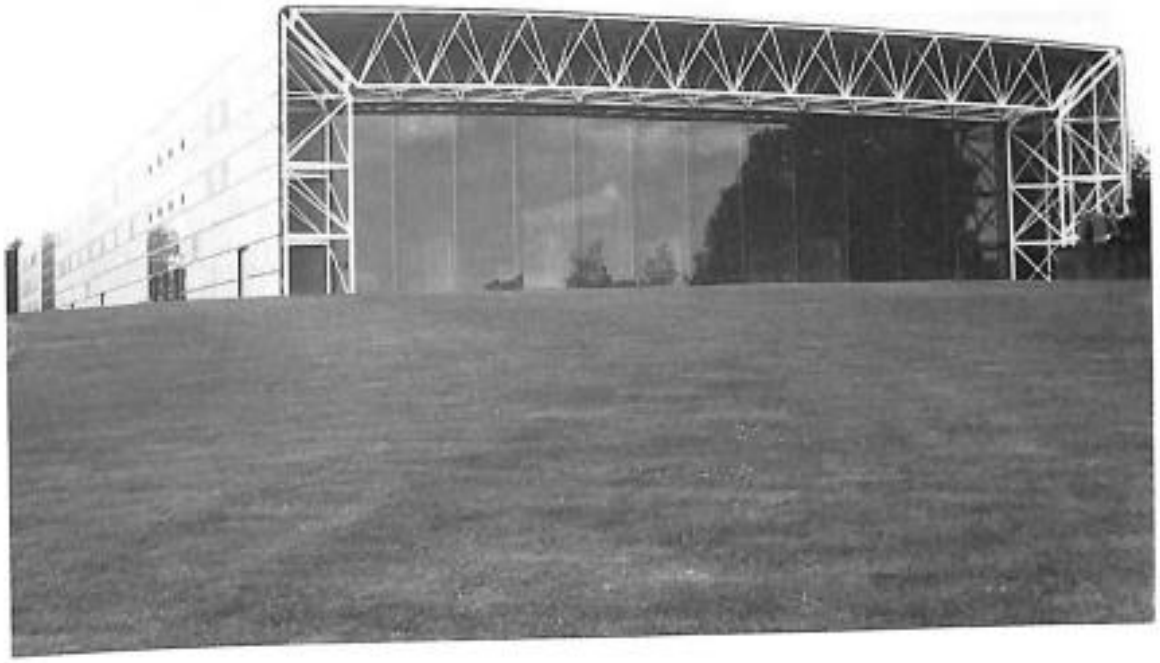
Description:
Single-storey art gallery with lattice steel columns and roof trusses and box-type panel cladding
Architects:
Foster Associates
Structural engineers:
A. Hunt Associates

Axonometric view from S:

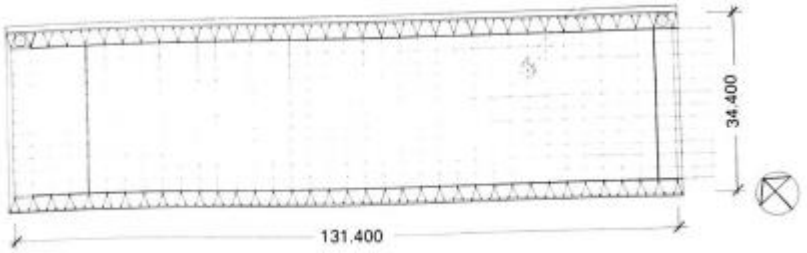


Steel Buildings: Complex Engineering Prob

Scheme 2:

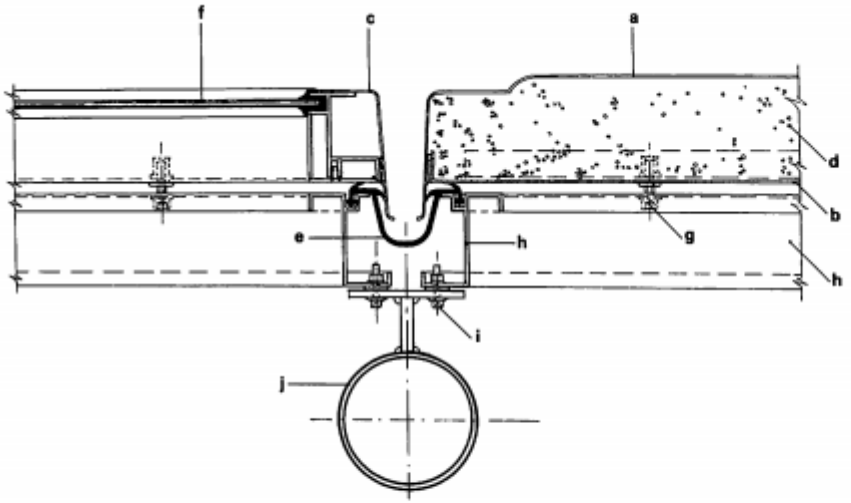


View of building from S.



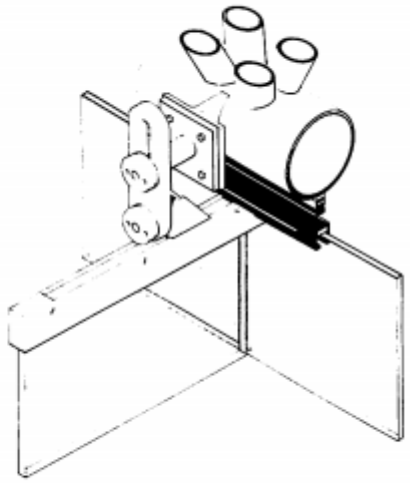
Steel Buildings: Complex Engineering Prob

Scheme 2:



a aluminium outer skin; b aluminium inner skin; c aluminium trim; d insulation; e neoprene gasket; f laminated glass; g stainless steel screws (6 per panel); h enamelled extruded aluminium subframe; i stainless steel nut and bolt; j outer tube of lattice truss/column.

Section on junction between glazed and solid panels.



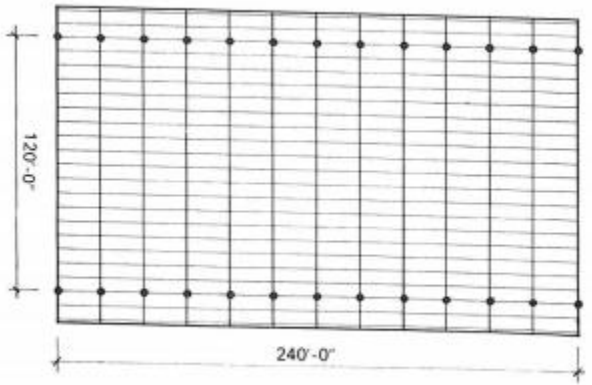
Junction between truss and glass gable wall (drawn by architect).

Steel Buildings: Complex Engineering Prob

Scheme 3:

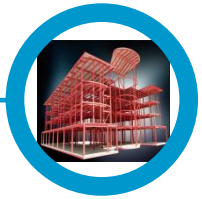


View of pavilion from N (photo by architect).



Plan at roof level.

Steel Buildings: Complex Engineering Prob

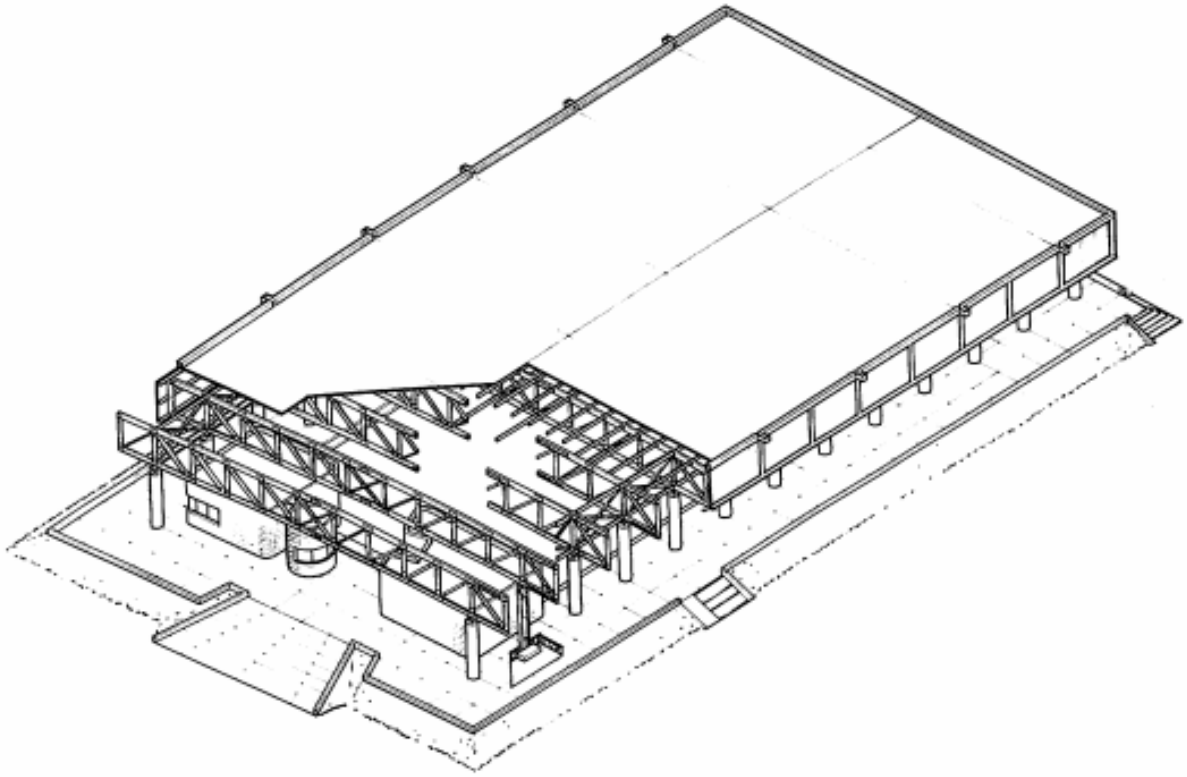


Scheme 3:

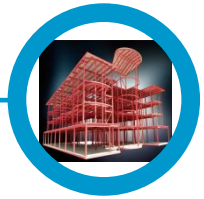
ANACOSTIA PARK PAVILION (1976)

Description:
Single-storey pavilion with precast concrete columns and timber roof trusses
Architects:
Keyes, Lethbridge and Condon
Structural engineer:
J. M. Cutts

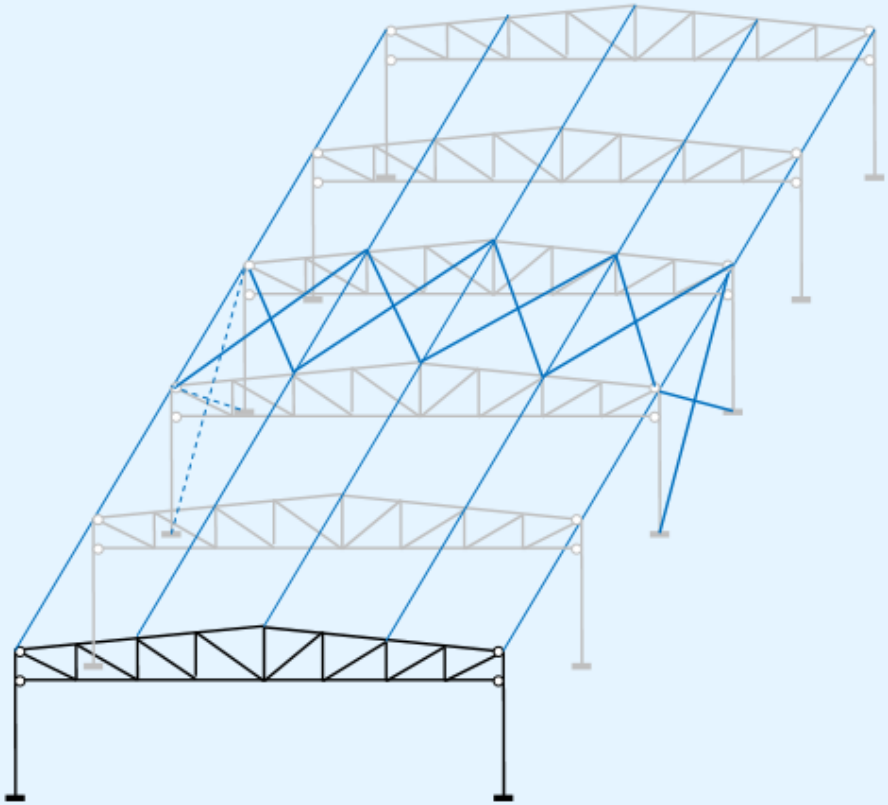
Axonometric view from E:



Steel Buildings: Complex Engineering Prob



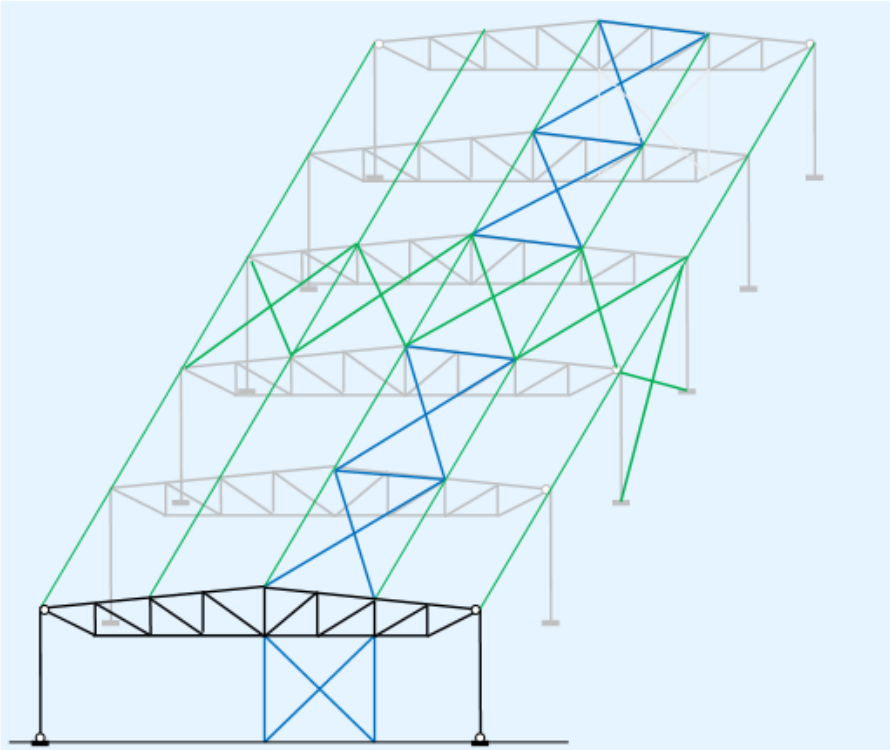
General



Lateral stability provided by portal trusses

Longitudinal stability provided by transverse wind girder and vertical cross bracings (blue)

No longitudinal wind girder



Vertical trusses are simply supported by columns

Lateral stability provided by longitudinal wind girder and vertical bracings in the gables (blue)

Longitudinal stability provided by transverse wind girder and vertical bracings (green)


Questions?